

# Quantum Metrology Enhanced by Leveraging Informative Noise with Error Correction

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The primary challenge in advancing practical quantum technology is the presence of noise, which can lead to decoherence and undermine the advantages of quantum systems. However, it is worth noting that noise can also contain information that can be harnessed to improve performance in certain quantum information tasks. This has been explored for specific types of noise, but the full potential of informative noise within general dynamics remains largely unknown. Here we demonstrate that informative noise can be transformed into a valuable resource for enhancing precision limits in quantum metrology. We provide necessary and sufficient conditions under which the precision achieved in the presence of informative noise can attain Heisenberg scaling. Our research offers a fresh perspective on the role of noise in quantum metrology and opens up avenues for achieving higher precision beyond the previously anticipated ultimate limit.

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**Introduction**—Quantum metrology, which makes use of quantum mechanical effects to improve the precision limit of parameter estimation [1–12], has gained increasing attention with a wide range of applications [13–33]. In practice, however, achieving the high precision promised by quantum metrology remains a challenging task. The main obstacle is the noise, which can destroy the potential advantages of almost all quantum technologies [5,15,34,35]. Current effort is typically trying to suppress the noise and restoring the dynamics to unitary evolutions [36–48] as it is believed that the unitary evolution leads to the highest precision. However, this belief is rooted in the assumption that parameters are solely encoded in the Hamiltonian. In practice, there are scenarios where both the Hamiltonian and the noise operators can encode the parameter, resulting in the noise itself becoming informative.

In this Letter, we study the precision limit of parameter estimation under general Markovian dynamics where the noise operators also contain the information of the parameter. We show that with proper controls such as quantum error correction (QEC), such informative noises can be valuable resources that can be exploited to improve the precision limit. We further provide necessary and sufficient conditions under which the precision achievable in the

presence of informative noise can reach the Heisenberg scaling, with significant enhancements attributed to the informative noise. Our study provides a new path in quantum metrology that can lead to higher precision limits by generalizing traditional schemes where only Hamiltonian is informative. It can have wide implications in practical applications, such as quantum gyroscope, quantum reference frame alignment, etc, where noises are in general unavoidable.

**Preliminaries**—We focus on single-parameter estimation where the precision limit of estimating a parameter,  $x$ , encoded in a quantum state,  $\rho_x$ , can be calibrated by the quantum Cramér-Rao bound (QCRB) [49–51] as  $\delta\hat{x} \geq 1/\sqrt{nJ_Q}$  with  $\delta\hat{x} = \sqrt{E[(\hat{x} - x)^2]}$  being the standard deviation of an unbiased estimator  $\hat{x}$  and  $n$  being the number of copies of the state. Here  $J_Q = \text{Tr}(\rho_x L_s^2)$  denotes the quantum Fisher information (QFI) for the parameter  $x$ ,  $L_s$  is the symmetric logarithmic derivative (SLD) which can be obtained from the equation  $\partial_x \rho_x = (L_s \rho_x + \rho_x L_s)/2$ . The QFI is closely related to the distance between two neighboring quantum states  $\rho_x$  and  $\rho_{x+dx}$  as [51]

$$d_{\text{Bures}}^2(\rho_x, \rho_{x+dx}) = \frac{1}{4} J_Q dx^2, \quad (1)$$

where  $d_{\text{Bures}}(\rho_x, \rho_{x+dx}) = \sqrt{2 - 2F(\rho_x, \rho_{x+dx})}$  is the Bures distance between  $\rho_x$  and  $\rho_{x+dx}$ ,  $F(\rho_1, \rho_2) = \text{Tr} \sqrt{\rho_1^{1/2} \rho_2 \rho_1^{1/2}}$

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is the fidelity of  $\rho_1$  and  $\rho_2$ . Intuitively the faster the state changes with the parameter, the higher the precision will be. Under unitary dynamics, the state evolves as  $|\psi(t)\rangle = U(t)|\psi(0)\rangle$  with  $dU(t)/dt = -iH(x)U(t)$ . In this case the QFI can be computed as

$$J_Q = 4\Delta h_x^2, \quad (2)$$

where  $h_x = \int_0^t U^\dagger(\tau) \partial_x H(x) U(\tau) d\tau$  [52–55] and  $\Delta h_x^2 = \langle \psi(0) | h_x^2 | \psi(0) \rangle - \langle \psi(0) | h_x | \psi(0) \rangle^2$  is the variance of  $h_x$  with respect to the initial probe state  $|\psi(0)\rangle$ .

In this Letter we consider the general sequential scheme that allows adaptive operations during the evolution, as illustrated in Fig. 1(b). The widely used parallel scheme, as shown in Fig. 1(a), is a special case of the general sequential scheme with the control operations taken as SWAP gates, which exchange the states of the system and different ancillas.

To illustrate the role of informative noises, we start with a toy model, where a spin is used to measure the direction of a fluctuating magnetic field. The dynamics can be described as

$$\frac{d|\psi\rangle}{dt} = -i[B + \xi(t)]\sigma_{\vec{n}}|\psi\rangle, \quad (3)$$

where  $\sigma_{\vec{n}} = \vec{n} \cdot \vec{\sigma}$  and  $\vec{n}$  denotes the direction of the field with respect to prefixed axes,  $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$  is the vector of Pauli matrices, and  $\xi(t)$  represents the fluctuation which can either arise from the fluctuation of the magnetic field itself or from the environment. We assume that the fluctuation is Markovian with a white spectrum,  $E[\xi(t)] = 0$ ,  $E[\xi(t)\xi(\tau)] = \gamma\delta(t - \tau)$ , which has the most detrimental effects. For simplicity, we consider a magnetic field in the XZ-plane, in which case  $\sigma_{\vec{n}(\theta)} = \cos\theta\sigma_1 + \sin\theta\sigma_3$  with  $\theta$  representing the direction of the magnetic field. The estimation of  $\theta$  thus corresponds to the orientation of the magnetic field.

We now show how the fluctuation,  $\xi(t)$ , can potentially enhance the precision of the estimation under the sequential scheme. For the estimation of  $\theta$ , taking one realization of the fluctuation (i.e., a quantum trajectory), we have  $\partial_\theta H = [B + \xi(\tau)](-\sin\theta\sigma_1 + \cos\theta\sigma_3)$  and

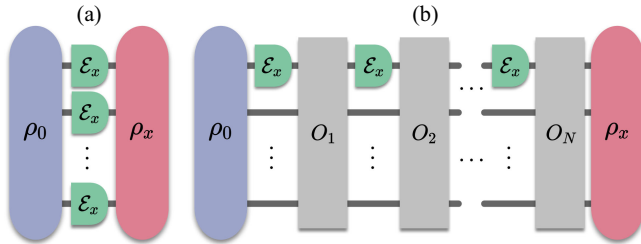


FIG. 1. Schemes for quantum metrology. (a) parallel scheme; (b) sequential scheme. Here  $\rho_0$  denotes the initial probe state,  $E_x$  denotes the dynamics,  $O_j$  denotes the control operations, and  $\rho_x$  denotes the output state.

$$h_\theta = \int_0^t U^\dagger(\tau)[B + \xi(\tau)](-\sin\theta\sigma_1 + \cos\theta\sigma_3)U(\tau)d\tau. \quad (4)$$

Assuming we can take a similar control strategy in the unitary dynamics [56,57] by adding controls to effectively make  $U(\tau) = I$ , then along one trajectory we have  $h_\theta = [Bt + \int_0^t \xi(\tau)d\tau](-\sin\theta\sigma_1 + \cos\theta\sigma_3)$ . The maximal variance with the optimal probe state is then  $\Delta h_\theta^2 = (Bt + \int_0^t \xi(\tau)d\tau)^2$ . Since the QFI is a statistical quantity which is only meaningful with sufficient repetitions and for different repetitions we have different realization of the fluctuation, the averaged maximal variance of  $h_\theta$  is thus given by  $E[\Delta h_\theta^2] = E[(Bt + \int_0^t \xi(\tau)d\tau)^2] = B^2t^2 + \gamma t$ . The corresponding QFI is then

$$J_Q = 4E[\Delta h_\theta^2] = 4B^2t^2 + 4\gamma t. \quad (5)$$

In addition to the Heisenberg scaling term,  $4B^2t^2$ , attributed to the ideal magnetic field  $B\sigma_{\vec{n}}$ , an additional term,  $4\gamma t$ , appears as an enhancement resulting from the informative noise  $\xi(t)\sigma_{\vec{n}}$ .

The positive effect of the fluctuation  $\xi(t)$  can also be seen from the effective master equation, which is

$$\frac{d\rho}{dt} = -i[B\sigma_{\vec{n}(\theta)}, \rho] + \gamma[\sigma_{\vec{n}(\theta)}\rho\sigma_{\vec{n}(\theta)} - \rho]. \quad (6)$$

We can immediately see that the parameter  $\theta$  is encoded in both the Hamiltonian and the noise operators. In such correlated parametrization, in addition to the Hamiltonian, noise operators can also contribute to the precision, which makes it possible to achieve higher precision.

To reap this improvement we need to design controls that make  $U(\tau) = I$ ; this requires tracking each trajectory precisely in order to reverse it. A natural question is what are the explicit strategies that can achieve this improvement? Furthermore, it is not clear whether this improvement is maximal. Can we identify the maximal improvement and construct protocols to achieve it?

*Noise-enhanced precision limit*—We study these questions for evolution with informative noises under general Markovian dynamics. The general Markovian dynamics where both the Hamiltonian and noisy operators containing the parameter can be written as

$$\begin{aligned} \frac{d\rho}{dt} = & -i[H(x), \rho] \\ & + \sum_{k=1}^m \left[ E_k(x)\rho E_k^\dagger(x) - \frac{1}{2}\{E_k^\dagger(x)E_k(x), \rho\} \right], \end{aligned} \quad (7)$$

where  $\{E_k(x)|k = 1, \dots, m\}$  are the Lindblad operators,  $x$  is the parameter to be estimated. We adopt the sequential scheme as depicted in Fig. 1(b). As the main result of this Letter, we present the following theorem.

*Theorem 1*—For general Markovian dynamics described by Eq. (7), the Heisenberg scaling can be achieved if and only if

$$\hat{H} = \dot{H} + \frac{i}{2} \sum_k (E_k^\dagger \dot{E}_k - \dot{E}_k^\dagger E_k) \notin \mathcal{S}, \quad (8)$$

where  $\mathcal{S} = \text{span}\{I, E_k, E_k^\dagger, E_k^\dagger E_j\}$  and the over dot refers to the differentiation with respect to  $x$ .

Furthermore, when the condition in Eq. (8) is satisfied, we can construct an explicit adaptive QEC strategy to achieve the QFI as

$$J_Q = 4\Delta_{|\psi\rangle}^2(\tilde{H})t^2 + 4\Gamma_{|\psi\rangle}(H, E_k)t, \quad \Gamma_{|\psi\rangle}(H, E_k) > 0, \quad (9)$$

where  $\Delta_{|\psi\rangle}^2(\tilde{H}) = \langle \psi | \tilde{H}^2 | \psi \rangle - \langle \psi | \tilde{H} | \psi \rangle^2$ . Here  $|\psi\rangle$  is the initial state and  $\tilde{H} = \Pi_C \hat{H} \Pi_C$  is always nontrivial with  $\Pi_C$  as the projection on the code space  $C$ .  $\Gamma_{|\psi\rangle}(H, E_k)$  is a positive quantity.

The influence of informative noise is clearly manifested in  $\Gamma_{|\psi\rangle}(H, E_k)$  and the additional terms in  $\hat{H}$ . The informative noise can thus serve as a resource which improves the precision limit.

We leave the proof of necessity in Sec. I of [58], which is analogous to previous studies [47,48]. To prove the sufficiency, we provide an explicit strategy that utilizes adaptive QEC to achieve the QFI in Eq. (9). Different from previous studies where the purpose of the QEC is to eliminate the noises and restore the noisy dynamics to the unitary evolution completely (the precision is thus at best the same as those under the unitary evolution) [36–42,47,48], with the correlated parametrization the aim of the adaptive QEC is to eliminate the negative effects of the noises but keep the positive contributions of the noises. We now show how this can be achieved with the adaptive QEC.

We assume the QEC can be implemented in a time scale much faster than the dynamics, thus at each time point a code space can be designed adaptively which satisfies the QEC conditions

$$\begin{aligned} \lim_{\hat{x} \rightarrow x} \Pi_C(\hat{x}) E_k(x) \Pi_C(\hat{x}) &= \alpha_k \Pi_C(\hat{x}), \\ \lim_{\hat{x} \rightarrow x} \Pi_C(\hat{x}) E_k^\dagger(x) E_j(x) \Pi_C(\hat{x}) &= \beta_{kj}(t) \Pi_C(\hat{x}). \end{aligned} \quad (10)$$

Here  $\Pi_C(\hat{x})$  denotes the projection on the code space designed according to the estimated value of the parameter  $\hat{x}$ . We now derive the precision limit achievable in the asymptotic limit, i.e., when  $\hat{x} \rightarrow x$ . Analysis for non-asymptotic case where  $\hat{x} \neq x$  are provided in [58].

Suppose at time  $t$ , the probe state is in the code space,  $\rho_C(t) = \Pi_C(\hat{x}) \rho_C(t) \Pi_C(\hat{x})$  [from now on we will also use  $\Pi_C$  as a short notation for  $\Pi_C(\hat{x})$  unless it is necessary to distinguish  $\hat{x}$  from  $x$ ], after a small  $dt$ , the states evolves to

$\rho(t+dt)$ . A syndrome measurement with projections  $\{\Pi_C, I - \Pi_C\}$  is then performed on the state, which gives

$$\begin{aligned} \rho_p(t+dt) &= \Pi_C \rho(t+dt) \Pi_C + \sum_{k=1}^m M_k \rho(t) M_k^\dagger \\ &= \Pi_C \rho(t+dt) \Pi_C + \sum_{k=1}^m d_{kk} U_k \Pi_C \rho(t) \Pi_C U_k^\dagger dt, \end{aligned} \quad (11)$$

where  $M_k(x) = [I - \Pi_C] E_k(x) \Pi_C \sqrt{dt}$  describes the noisy effects that drive the state out of the code space (see Sec. II of [58] for details). The recovery operation,  $R_E$ , can be taken as the operation with the Kraus operators consisting of  $\{\Pi_C, \Pi_C U_1^\dagger(\hat{x}), \dots, \Pi_C U_m^\dagger(\hat{x})\}$ , where  $\hat{U}_k^\dagger(\hat{x})$  is obtained from the estimated error operator  $M_k(\hat{x}) = (I - \Pi_C(\hat{x})) E_k(\hat{x}) \Pi_C(\hat{x}) \sqrt{dt} = \sqrt{d_{kk}} \hat{U}_k(\hat{x}) \Pi_C(\hat{x}) \sqrt{dt}$ . The recovered state at  $t+dt$  is then given by  $\rho_C(t+dt) = R_E[\rho_p(t+dt)]$ . From the difference between  $\rho_C(t+dt)$  and  $\rho_C(t)$ , we can then obtain the dynamics for the recovered state,  $d\rho_C/dt = L_x^C(\rho_C)$ . In Sec. II of [58] we show that the effective dynamics  $L_x^C$  can be expanded around  $\hat{x}$  as

$$\frac{d\rho_C}{dt} = [L_1 dx + L_2 dx^2 + O(dx^3)](\rho_C), \quad (12)$$

with  $dx = x - \hat{x}$ ,

$$\begin{aligned} L_1(\rho) &= -i \left[ \Pi_C \left( \dot{H} + \frac{i}{2} \sum_k E_k^\dagger \dot{E}_k - \dot{E}_k^\dagger E_k \right) \Pi_C, \rho \right], \\ L_2(\rho) &= -i \frac{1}{2} \left[ \Pi_C \left( \ddot{H} + \frac{i}{2} \sum_k E_k^\dagger \ddot{E}_k - \ddot{E}_k^\dagger E_k \right) \Pi_C, \rho \right] \\ &\quad + \sum_k \left[ \Pi_C \dot{E}_k \Pi_C \rho \Pi_C \dot{E}_k^\dagger \Pi_C - \frac{1}{2} \{ \Pi_C \dot{E}_k^\dagger \dot{E}_k \Pi_C, \rho \} \right] \\ &\quad + \sum_{k,j} \frac{1}{d_{kk}} \Pi_C (E_k^\dagger \dot{E}_j - \alpha_k^* \dot{E}_j) \Pi_C \rho \Pi_C (\dot{E}_j^\dagger E_k - \alpha_k \dot{E}_j^\dagger) \Pi_C, \end{aligned} \quad (13)$$

where we use the over dot to denote the derivatives with respect to  $x$ , which is then evaluated at the point  $x = \hat{x}$ , for example,  $\ddot{H} = \partial_x^2 H|_{x=\hat{x}}$ .

The second order expansion of  $dx$  is sufficient for the computation of QFI since from Eq. (1) we know the QFI is determined by the distance between two neighboring states,  $\rho_C(x)$  and  $\rho_C(x+dx)$ , up to the second order of  $dx$ . In the asymptotic limit when  $\hat{x} = x$ ,  $\rho_C(x)$  can be obtained from Eq. (12) with  $dx = 0$ , which gives  $\rho_C(x) = \rho_0$ . While  $\rho_C(x+dx)$ , up to the second order of  $dx$ , can be obtained as  $\rho_C(x+dx) = e^{(L_1 dx + L_2 dx^2)t}(\rho_0) = [I + tL_1 dx + tL_2 dx^2 + \frac{1}{2} t^2 L_1^2 dx^2 + O(dx^3)](\rho_0)$ . If we take the initial

probe state as  $\rho_0 = |\psi\rangle\langle\psi|$ , the fidelity between  $\rho_C(x)$  and  $\rho_C(x+dx)$  is given as

$$\begin{aligned} F^2(\rho_x, \rho_{x+dx}) &= \langle\psi|\rho_C(x+dx)|\psi\rangle \\ &= 1 + tdx\langle\psi|L_1(|\psi\rangle\langle\psi|)|\psi\rangle \\ &\quad + \frac{1}{2}t^2dx^2\langle\psi|L_1^2(|\psi\rangle\langle\psi|)|\psi\rangle \\ &\quad + tdx^2\langle\psi|L_2(|\psi\rangle\langle\psi|)|\psi\rangle + O(dx^3). \end{aligned} \quad (14)$$

Using Eq. (13), it is easy to obtain

$$\begin{aligned} \langle\psi|L_1(|\psi\rangle\langle\psi|)|\psi\rangle &= 0, \\ \langle\psi|L_1^2(|\psi\rangle\langle\psi|)|\psi\rangle &= -2[\langle\psi|\tilde{H}^2|\psi\rangle - \langle\psi|\tilde{H}|\psi\rangle^2], \end{aligned} \quad (15)$$

where  $\tilde{H} = \Pi_C[\hat{H} + (i/2)\sum_k(E_k^\dagger\dot{E}_k - \dot{E}_k^\dagger E_k)]\Pi_C$ . From Eq. (1) we can then obtain the QFI as

$$\begin{aligned} J_Q &= \lim_{dx \rightarrow 0} \frac{8 - 8F[\rho_C(x), \rho_C(x+dx)]}{dx^2} \\ &= 4\Delta_{|\psi\rangle}^2(\tilde{H})t^2 + 4\Gamma_{|\psi\rangle}(H, E_k)t, \end{aligned} \quad (16)$$

where  $\Delta_{|\psi\rangle}^2(\tilde{H}) = \langle\psi|\tilde{H}^2|\psi\rangle - \langle\psi|\tilde{H}|\psi\rangle^2$ ,  $\Gamma_{|\psi\rangle}(H, E_k) = -\langle\psi|L_2(|\psi\rangle\langle\psi|)|\psi\rangle$ .

This concludes the proof of sufficiency for Theorem 1.

The constructive protocol can be succinctly outlined in the following steps: (i) Taking the initial state  $|\psi\rangle$  as a pure state in the code space. In the long time limit where the first term in Eq. (9) dominates, the optimal initial state in code space  $C$  is  $(|C_{\min}\rangle + |C_{\max}\rangle)/\sqrt{2}$ , where  $|C_{\min}\rangle$  and  $|C_{\max}\rangle$  are the eigenstates of  $\tilde{H}$  corresponding to the minimal and maximal eigenvalues. The method for designing the optimal code space is similar to that of previous studies [48], with the key difference being that the effective Hamiltonian is replaced by  $\tilde{H}$ . (ii) Repetitively apply the recovery operations with Kraus operators consisting of  $\{\Pi_C, \Pi_C U_1^\dagger(\hat{x}), \dots, \Pi_C U_m^\dagger(\hat{x})\}$ , where  $\hat{U}_k^\dagger(\hat{x})$  is obtained with polar decomposition from the estimated error operator,  $M_k(\hat{x}) = [I - \Pi_C(\hat{x})]E_k(\hat{x})\Pi_C(\hat{x})\sqrt{dt} = \sqrt{d_{kk}}\hat{U}_k(\hat{x}) \times \Pi_C(\hat{x})\sqrt{dt}$ . (iii) Perform projective measurement on the eigenvectors of the SLD operator of the final state.

We note that Eq. (8) is similar to the conventional schemes [48] but with additional terms contributed from the noises. In Sec. II of [58] we show that  $\Gamma_{|\psi\rangle}(H, E_k)$  is always nonnegative, indicating a significant improvement in the QFI attributed to the Lindblad operators. The additional terms in  $\tilde{H}$  contributed by the Lindblad operators may also enhance the prefactor of the Heisenberg scaling  $t^2$ . The noise thus serves as a resource which can improve the precision limit (see Fig. 2 for a geometrical interpretation).

*Toy model revisited*—We now apply this general result to the toy model. For the estimation of  $\theta$  with dynamics given

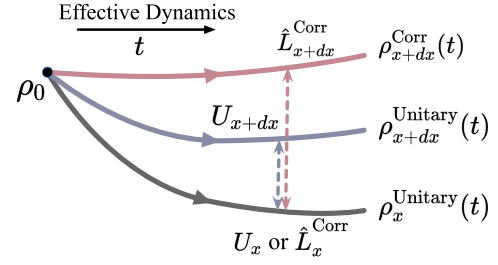


FIG. 2. Illustration of how correlated parametrization with adaptive QEC can lead to more diverging dynamics which improves the precision. Under the correlated parametrization, the adaptive QEC only corrects the noise at one particular value of the parameter  $x$ , and at this particular value the corrected dynamics,  $\hat{L}_x^{\text{Corr}}$ , becomes unitary. However, the neighboring dynamics, denoted as  $\hat{L}_{x+dx}^{\text{Corr}}$ , still contains the noise. Compared to the purely unitary dynamics the noise makes  $\hat{L}_{x+dx}^{\text{Corr}}$  more different from  $\hat{L}_x^{\text{Corr}}$ , which makes the neighboring states further apart (quantified by the Bures distance). This makes the correlated parametrization outperform the unitary evolution.

by Eq. (6), we have  $H(\theta) = B\sigma_{\vec{n}(\theta)} = B(\cos\theta\sigma_1 + \sin\theta\sigma_3)$ ,  $E = \sqrt{\gamma}\sigma_{\vec{n}(\theta)} = \sqrt{\gamma}(\cos\theta\sigma_1 + \sin\theta\sigma_3)$ . Thus

$$\begin{aligned} \hat{H} &= \partial_\theta H + \frac{i}{2}[E^\dagger \partial_\theta E - (\partial_\theta E^\dagger)E] \\ &= B(-\sin\theta\sigma_1 + \cos\theta\sigma_3) + \gamma\sigma_2 \notin \text{span}\{I, E\}, \end{aligned} \quad (17)$$

and the Heisenberg scaling can be achieved. By employing an ancillary qubit, we can choose the code space as  $\{|C_0\rangle = |+\hat{H}\rangle|0\rangle, |C_1\rangle = |-\hat{H}\rangle|1\rangle\}$ , where  $|\pm\hat{H}\rangle$  are the eigenvectors of  $\hat{H}$ . In this case  $\tilde{H} = \Pi_C \hat{H} \Pi_C = \hat{H}$ . From Eq. (13) we have

$$\begin{aligned} L_1(\rho) &= -i[B(-\sin\theta\sigma_1 + \cos\theta\sigma_3) + \gamma\sigma_2, \rho], \\ L_2(\rho) &= \gamma(\sigma_3^C \rho \sigma_3^C - \rho), \end{aligned} \quad (18)$$

where  $\sigma_3^C = |C_0\rangle\langle C_0| - |C_1\rangle\langle C_1|$ . If we take the initial probe state as  $|\psi\rangle = (|C_0\rangle + |C_1\rangle)/\sqrt{2}$ , then the QFI can be obtained directly as

$$J_Q = 4(B^2 + \gamma^2)t^2 + 4\gamma t. \quad (19)$$

While without the informative noise ( $\gamma = 0$ ), the highest QFI is  $J_Q = 4B^2t^2$ . We thus obtain an improvement on the prefactor of the Heisenberg scaling, and the total improvement, which is  $1 + \gamma^2/B^2$  in the asymptotic limit of  $t \rightarrow \infty$ , increases with the strength of the noise. In Sec. III of [58], we present a detailed design of the QEC protocol using this code. This includes the initial state setup, recovery operations, and the optimal measurement strategy. Additionally, we include a numerical simulation that demonstrates the practical improvements in realistic scenarios.

It can be seen that the QFI obtained previously,  $J_Q = 4B^2t^2 + 4\gamma t$ , is suboptimal. This corresponds to a suboptimal error correction code. Specifically, this corresponds to choosing the code space with the basis taken as  $|\tilde{C}_0\rangle = |+\partial_\theta H\rangle|0\rangle$  and  $|\tilde{C}_1\rangle = |-\partial_\theta H\rangle|1\rangle$ , where  $|\pm\partial_\theta H\rangle$  are the eigenvectors of  $\partial_\theta H = B(-\sin\theta\sigma_1 + \cos\theta\sigma_3)$ . In this case  $\tilde{H} = \Pi_C \hat{H} \Pi_C = \partial_\theta H$ . When the initial probe state taken as  $|\psi\rangle = (|\tilde{C}_0\rangle + |\tilde{C}_1\rangle)/\sqrt{2}$ , the QFI is then exactly

$$J_Q = 4B^2t^2 + 4\gamma t. \quad (20)$$

$\{|\tilde{C}_0\rangle, |\tilde{C}_1\rangle\}$  is thus a suboptimal QEC code for this model.

In Sec. IV of [58] we show the analysis can be easily extended to the time-dependent master equation

$$\begin{aligned} \frac{d\rho}{dt} = & -i[H(x, t), \rho] \\ & + \sum_{k=1}^m \left[ E_k(x, t) \rho E_k^\dagger(x, t) - \frac{1}{2} \{E_k^\dagger(x, t) E_k(x, t), \rho\} \right], \end{aligned} \quad (21)$$

where  $H(x, t)$  and  $\{E_k(x, t)\}$  are time dependent. This can be applied to estimating the frequency of a rotating field where the dynamic is given by

$$\frac{d\rho}{dt} = -i[B\sigma_{\bar{n}}(\Omega, t), \rho] + \gamma[\sigma_{\bar{n}}(\Omega, t)\rho\sigma_{\bar{n}}(\Omega, t) - \rho], \quad (22)$$

where  $\sigma_{\bar{n}}(\Omega, t) = \cos(\Omega t)\sigma_1 - \sin(\Omega t)\sigma_2$  and  $\Omega$  is the frequency to be estimated. It is shown that in this case the QFI can reach  $(B^2 + \gamma^2)t^4 + \frac{4}{3}\gamma t^3$ . Significant enhancements attributed to the informative noise thus still exists. The details can be found in [58], including a numerical simulation demonstrating the practical improvement.

**Summary**—Our study provides a new perspective on the role of noise in quantum metrology. We demonstrate that informative noise in quantum dynamics can be reshaped to improve precision in various quantum metrology tasks. This deepens our understanding of the ultimate achievable precision in quantum metrology. Our study suggests that instead of solely attempting to suppress noise, in cases where noise is informative due to correlated parametrization, increasing the noise can actually be beneficial.

It is important to note that our approach differs from previous studies in quantum computation that utilize noise, such as environmental engineering and noise-assisted quantum error correction. In those cases, explicit control or design of the environment is required, and typically the performance cannot surpass that of unitary dynamics.

We anticipate that our study will stimulate exploration of correlated parametrization in numerous applications and uncover new limits to ultimate precision that were previously believed to be unattainable. Future studies could include extending our findings to multiparameter quantum

estimation and intentionally engineering correlated parametrization for various applications.

*Note added*—Recently, a paper [60] appeared, where a condition similar to Eq. (8) is given. This provides an alternative proof for the necessity of Theorem 1 in this Letter.

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